IPL Auction Decision System

A Non-Parametric and Decision-Theoretic Approach

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1 Introduction

This report presents an IPL Auction Decision System built using ball-by-ball cricket data. The model estimates player value and determines an optimal bidding strategy using statistical modeling, simulation, and decision theory.

The dataset consists of detailed match-level observations aggregated into player-level statistics such as runs, strike rate, wickets, and economy.

2 Data Aggregation and Feature Construction

The raw ball-by-ball data is transformed into meaningful player statistics:

- Batting metrics: Runs, Strike Rate (SR), Average (Avg)
- Bowling metrics: Wickets, Economy Rate
- Role classification: Batsman, Bowler, All-rounder

These features summarize player performance over multiple seasons.

3 Value Function and Justification

3.1 Why Normalization?

Different variables have different scales:

- Runs: up to 8652
- Wickets: up to 221
- Economy: up to 36

Without normalization, variables with larger magnitude dominate the model.
Thus, Min-Max normalization is applied:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

3.2 Regression-Based Value Function

The total player value is computed as:

$$V = \beta_0 + \beta_1 Runs_n + \beta_2 SR_n + \beta_3 Avg_n + \beta_4 Wkts_n + \beta_5 Econ_n \tag{2}$$

Where coefficients are obtained using regression on auction data.
Reason:

- Captures marginal contribution of each feature
- Data-driven (not arbitrary weights)
- Interpretable economic meaning (Rs per unit increase)

4 Uncertainty Modeling

Player value is uncertain due to:

- Match conditions
- Team demand
- Player form

Thus, value is modeled as:

$$V \sim \mathcal{N}(\mu, \sigma^2) \tag{3}$$

where:

$$\sigma = 0.3 \times \mu \tag{4}$$

Why Normal Distribution?

- Central Limit Theorem justification
- Captures symmetric uncertainty
- Easy simulation

5 Optimal Bidding Strategy

The objective is to maximize expected utility:

$$U(b) = P(\text{win} \mid b) \cdot E[V - b \mid \text{win}] - \lambda b \tag{5}$$

Where:

- b = bid amount
- λ = risk aversion

Interpretation:

- Higher bid increases win probability

- But reduces profit margin
- Risk term penalizes aggressive bidding

Monte Carlo simulation is used to estimate expectations.

6 Limitations of the Model

- Assumes normal distribution (may not hold in real auctions)
- Fixed uncertainty (30%) for all players
- Ignores contextual factors:
  - Team composition
  - Pitch conditions
  - Player popularity
- Regression trained on limited sample (38 players)
- Linear model cannot capture complex interactions

7 Possible Improvements

- Use non-linear models (Random Forest, Neural Networks)
- Player clustering using unsupervised learning
- Dynamic uncertainty estimation
- Bayesian updating with new performance data
- Incorporate match context and team needs

8 Connection to Non-Parametric Methods

This model relates to Non-Parametric Inference in the following ways:

- No strict assumption on underlying distribution of player performance
- Empirical aggregation of data instead of parametric modeling
- Monte Carlo simulation approximates unknown distributions

Possible Non-Parametric Extensions:

- Kernel Density Estimation (KDE) for player value distribution
- Bootstrap methods for uncertainty estimation
- Rank-based evaluation instead of mean-based metrics

9 Connection to Decision Theory

The model is fundamentally a decision-theoretic framework:

- Objective: Maximize expected utility
- Trade-off: Risk vs reward
- Action space: possible bids
- Uncertainty: player value distribution

Key Concepts Used:

- Expected Utility Maximization
- Risk Aversion
- Probabilistic decision making
- Simulation-based optimization

10 Conclusion

The IPL Auction Decision System provides a structured way to evaluate players and determine optimal bids using statistical modeling and decision theory. While the model simplifies real-world complexity, it serves as a strong foundation for data-driven sports analytics and auction strategy.